

Shuai WANG

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EDUCATION

- **City University of Hong Kong** Hong Kong, China
Ph.D. in Computer Science, supervised by Prof. Li Shuai Cheng
Thesis title: Computational Methods for Strain-Level Microbiota Characterization.
Aug 2018 – Oct 2022
- **Northwestern Polytechnical University** Xi An, China
Bachelor of Biotechnology, supervised by Prof. Qian Airong
Thesis title: The establishment of the mammalian RNA database.
Sep 2013 – Jul 2017

WORK EXPERIENCE

- **Research Scientist** Hong Kong, China
OmicLab limited, Hong Kong Science and Technology Park
Oct 2022 – Current
- **Bioinformatics Analyst** Shenzhen, China
Beijing Genomics Institute
Jul 2017 – Aug 2018

RESEARCH INTERESTS

- **Distinguish bacterial strains in microbiota:** Human microbiota is highly associated with host health, and it has the potential to serve as diagnosis biomarkers or even treatment targets for human diseases. Different strains of the same species may harbor distinct or even opposite associations with human health. However, existing widely used microbiota analysis methods are insufficient to study bacterial strains. I am interested in developing bioinformatics toolboxes to profile bacterial strains in the complex metagenomics data by incorporating the short- and long-read sequencing data, genotype frequency, population information, the database, *etc.* Specifically, I intend to develop methods to identify genomic variations such as structural variation (SV), and horizontal gene transfer (HGT), and then phase or assign the variants to distinguish bacterial strains.
- **Metagenomics assembly at the strain level:** Metagenomics assembly based on high throughput short read sequencing is usually highly fragmented. The development of sequencing technologies such as long read sequencing, Hi-C, linked-reads sequencing allows high quality metagenomics assemblies. I am interested in assembling the microbial strains directly by integrating the advanced sequencing technologies and additional information including genotype frequency, population information, the database, *etc.*

RESEARCH PROJECTS

- **Marker gene-based strain profiling:** Traditional strain profiling methods only phase the variants by the genotype frequency at variant loci. We developed a new iterative algorithm to incorporate the variant linkage information from both genotype frequency and sequencing reads to phase SNP at marker genes of each bacteria species.
- **Full-resolution HLA typing:** Use fine-tuned reads binning and local assembly to achieve accurate reads alignment in the HLA loci, and reconstruct full-length HLA sequences based on variant phasing. Integrate phase information from different sequencing technologies, genotype frequency, and pedigree data to advance the performance.
- **Ultrafast HGT identification from metagenomics data:** The bottleneck of HGT detection is to align the reads to the large reference database. We use approximate k-mer matching to rapidly select the reference segments that may contain HGT breakpoints to accelerate the reads alignment. Subsequently, we identify HGT breakpoint positions with discordant reads using a method similar to traditional SV detection.
- **Reference-based isolate bacteria assembly:** Currently, the NCBI database harbors plenty of bacteria assemblies. Given a isolate sequencing sample, we first identify its closest assembly from the database using kmer, and then align reads to the closet assembly to split the assembly into segments. Finally, we link the segments into a whole genome sequence of the isolate bacteria using bipartite matching.

RESEARCH PUBLICATIONS

- **Wang, S.,** Wang, M., Chen, L., ... and Li, S. (2023). SpecHLA: full-resolution HLA typing from sequencing data. *Cell Reports Methods*. Accepted.
- **Wang, S.,** Jiang, Y., and Li, S. (2021). PStrain: an iterative microbial strains profiling algorithm for shotgun metagenomic sequencing data. *Bioinformatics*, 36(22-23), 5499-5506.
- **Wang, S.,** Jiang, Y., and Li, S. (2023). Horizontal gene transfer events in the gut microbiome can be biomarkers for host health. In preparation.
- Jiang, Y., **Wang, S.,** Zhang X., and Li, S. (2021). A framework to trace microbial engraftment at the strain level during FMT. *International Conference on Intelligent Biology and Medicine (ICIBM)*, August 8-10, 2021, Virtual. (**Co-first author; Best paper award**)

- Wang, C., Li, H., Guo, Y., Huang, J., Sun, Y., Min, J., Wang, J., Fang, X., Zhao, Z., **Wang, S.**, ... Li, S., Zhou, X., and Zhong, J. (2020). Donkey genomes provide new insights into domestication and selection for coat color. *Nature communications*, 11(1), 1-15.
- Zhao, Z., Zhou, Y., **Wang, S.**, Zhang, X., Wang, C., and Li, S. (2020). LDscaff: LD-based scaffolding of de novo genome assemblies. *BMC bioinformatics*, 21(21), 1-15.

HONORS AND AWARDS

- Best paper award of the International Conference on Intelligent Biology and Medicine (ICIBM) in Aug 2021.
- BGI Outstanding Talent Scholarship from Beijing Genomics Institute College in Jan 2017.

SKILLS SUMMARY

- **Programming Languages:** Python, C++, Perl, R, Linux, Shell, $\text{\LaTeX}$
- **Bioinformatics:** Familiarity with multiple sequencing protocols; Proficient in standard pipelines such as alignment, variant detection, variant phasing, and assembly; Proficient in developing bioinformatics softwares.

DEVELOPED SOFTWARES

- **PStrain:** Phase SNPs of marker genes to distinguish bacteria strains from metagenomics data (Python).
- **SpecHLA:** Reconstruct HLA gene sequences from multiple data types (Python).
- **LocalHGT:** Use k-mer to rapidly detect HGT events from metagenomics data (Python/C++).